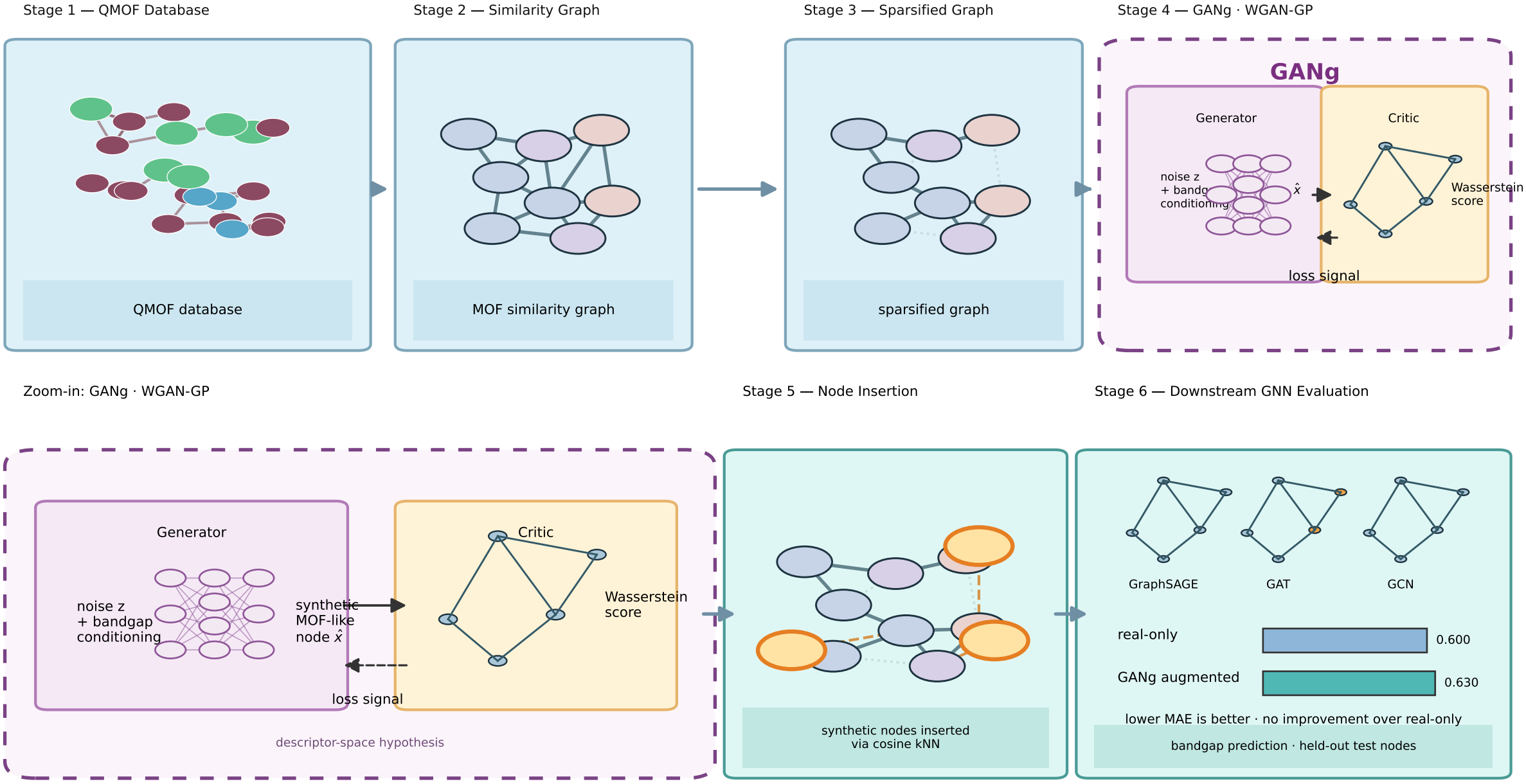
Stage 1 — QMOF Database
Stage 2 — Similarity Graph
Stage 3 — Sparsified Graph
Stage 4 — GANg · WGAN-GP
QMOF database
MOF similarity graph
sparsified graph
GANg
Generator
Critic
noise z
+ bandg
conditioning
$\hat{x}$
Wasserstein
score
loss signal
Zoom-in: GANg · WGAN-GP
Stage 5 — Node Insertion
Stage 6 — Downstream GNN Evaluation
Generator
noise z
+ bandgap
conditioning
synthetic
MOF-like
node $\hat{x}$
Critic
Wasserstein
score
loss signal
descriptor-space hypothesis
synthetic nodes inserted
via cosine kNN
GraphSAGE
GAT
GCN
real-only
0.600
GANg augmented
0.630
lower MAE is better · no improvement over real-only
bandgap prediction · held-out test nodes